



WIPRO NGA Program – 25SUB4508_LSP

Capstone Project Presentation – 13th Feb2026

Project Title Here - **Enterprise Concurrent Signal Broadcast Framework (ECSBF)**

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www.rpsconsulting.in

Introduction:

- A high-performance distributed communication framework for real-time signal propagation.
- Designed to enable secure and reliable communication across multiple networked nodes.
- Uses a centralized Engine to manage authentication, sessions, and broadcasting.
- Built with concurrency, fault tolerance, and encryption as core principles.
- Ensures synchronized, traceable, and secure enterprise communication.

PROBLEM STATEMENT:

- Distributed systems require secure and synchronized real-time communication.
- Managing multiple concurrent node connections is complex and error-prone.
- Unauthorized access can lead to signal injection and security breaches.
- Network failures may disrupt sessions and cause data inconsistency.
- Lack of centralized observability makes debugging and monitoring difficult

SOLUTION:

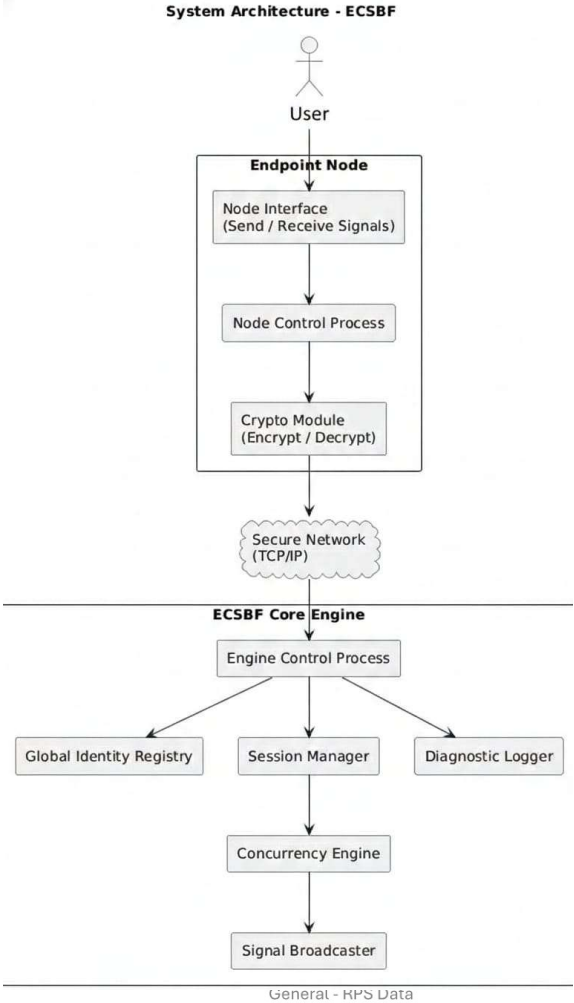
- Centralized ECSBF Engine manages all node communication securely.
- Global Identity Registry ensures only provisioned nodes can connect.
- Multi-threaded concurrency engine handles simultaneous signal requests.
- Encrypted signal broadcasting ensures confidentiality and integrity.
- Diagnostic logging provides observability and system transparency.

WORKFLOW:

- Endpoint Nodes connect to the ECSBF Engine via secure TCP/IP channels.
- Secure handshake verifies node identity before session establishment.
- Engine Control Process manages authentication and validation.
- Concurrency Engine processes signals from multiple nodes simultaneously.
- Signal Broadcaster distributes validated signals to all active nodes.
- Diagnostic Logger records system activities at multiple severity levels.

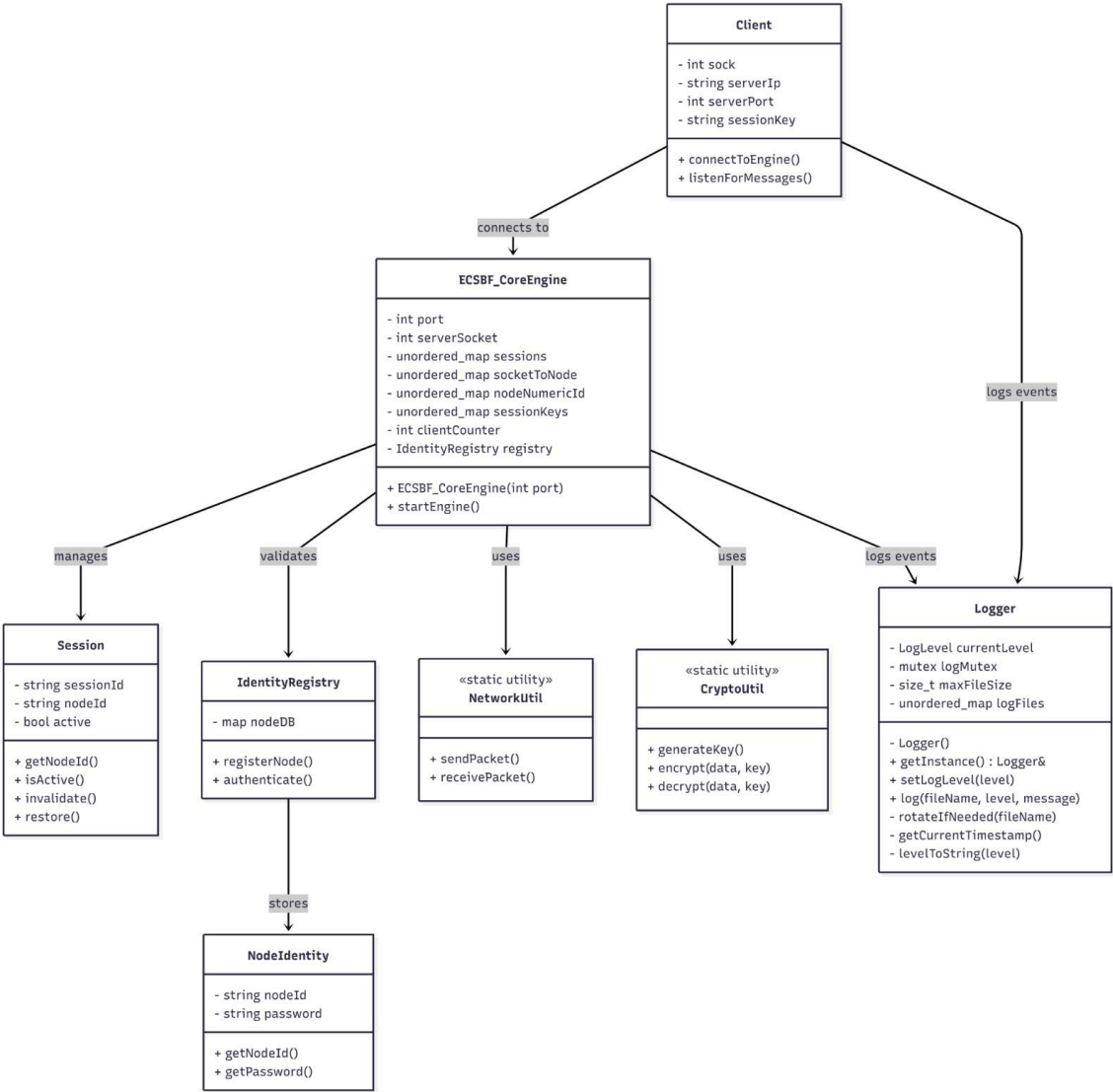
Enterprise Concurrent Signal Broadcast Framework (ECSBF)

SYSTEM ARCHITECTURE DIAGRAM:



Enterprise Concurrent Signal Broadcast Framework (ECSBF)

CLASS DIAGRAM:



PRACTICAL OUTCOMES – STEP 1 (Node Registration & Session Setup)

- Node initiates secure connection request to the Engine.
- Engine verifies identity from Global Identity Registry.
- Successful verification establishes a persistent session.
- Unauthorized nodes are rejected and logged securely.

Enterprise Concurrent Signal Broadcast Framework (ECSBF)

EXPLORER...

OPEN EDITORS

README.mdengine.log Xclient.log

engine.log logsclient.log logs

ECSBF...

.vscodeappsclientengineincludelogsclient.logengine.logsrcMakefileREADME.md

logs > engine.log

1 [2026-02-12 17:25:26] [INFO] Starting ECSBF Engine...
2 [2026-02-12 17:25:26] [INFO] Server socket created successfully
3 [2026-02-12 17:25:26] [INFO] Bind successful on port 4096
4 [2026-02-12 17:25:26] [INFO] Server now listening
5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

souvik@ray-device:/mnt/s/ECSBF_v1\$ make clean
Cleaning binaries...
rm -f apps/engine/engine apps/client/client
souvik@ray-device:/mnt/s/ECSBF_v1\$ make
Building ECSBF Engine...
g++ -std=c++17 -Wall -Iinclude -Iinclude/core -Iinclude/client -Iinclude/logging -Iinclude/network -Iinclude/registry -Iinclude/security apps/engine/main_engine.cpp src/core/ECSBF_CoreEngine.cpp src/session/Session.cpp src/logging/Logger.cpp src/network/NetworkUtil.cpp src/registry/IdentityRegistry.cpp src/security/NodeIdentity.cpp src/security/CryptoUtil.cpp -o apps/engine/engine
Building ECSBF Client...
g++ -std=c++17 -Wall -Iinclude -Iinclude/core -Iinclude/client -Iinclude/logging -Iinclude/network -Iinclude/registry -Iinclude/security apps/client/main_client.cpp src/client/Client.cpp src/logging/Logger.cpp src/network/NetworkUtil.cpp src/security/CryptoUtil.cpp -o apps/client/client
souvik@ray-device:/mnt/s/ECSBF_v1\$./apps/engine/engine

PRACTICAL OUTCOMES – STEP 2 (Signal Transmission & Broadcasting)

- Authenticated node sends encrypted signal payload to Engine.
- Engine validates active session before processing signal.
- Signal is tagged with source identifier for traceability.
- Broadcast mechanism sends signal to all active nodes concurrently.

Enterprise Concurrent Signal Broadcast Framework (ECSBF)

EXPLORER

...

README.mdengine.log Xclient.log

OPEN EDITORS

logs > engine.log

5 [2026-02-12 17:26:15] [INFO] New client connected

6 [2026-02-12 17:26:25] [INFO] New node registered: node1

7 [2026-02-12 17:26:25] [INFO] New client connected

8 [2026-02-12 17:26:45] [INFO] Node-1 connected

9 [2026-02-12 17:26:45] [INFO] Session key generated for Node-1

10 [2026-02-12 17:27:19] [INFO] New client connected

11 [2026-02-12 17:27:31] [INFO] New node registered: node2

12 [2026-02-12 17:27:31] [INFO] New client connected

13 [2026-02-12 17:27:52] [INFO] Node-2 connected

14 [2026-02-12 17:27:52] [INFO] Session key generated for Node-2

15

client.loglogs

ECSBF_V1

> .vscode

> apps

> client

> engine

> include

> logs

client.log

engine.log

node1.log

node2.log

> src

Makefile

README.md

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

souvik@ray-device:/mnt/s/ECSBF_v1\$./apps/client/client

===== ECSBF NODE =====

1. Register

2. Login

3. Exit

Choose option: 1

Enter UserID: node1

Enter Password: 123

REGISTER_SUCCESS

Please login to continue.

===== ECSBF NODE =====

1. Register

2. Login

3. Exit

Choose option: 2

Enter UserID: node1

Enter Password: 123

LOGIN_SUCCESS

You can now chat.

Type 'exit' to quit.

node1 hiii

Packet sent by Node-2: node2 hello

OUTLINE

TIMELINE

EXPLORER

...

README.mdengine.log Xclient.log

OPEN EDITORS

logs > engine.log

5 [2026-02-12 17:26:15] [INFO] New client connected

6 [2026-02-12 17:26:25] [INFO] New node registered: node1

7 [2026-02-12 17:26:25] [INFO] New client connected

8 [2026-02-12 17:26:45] [INFO] Node-1 connected

9 [2026-02-12 17:26:45] [INFO] Session key generated for Node-1

10 [2026-02-12 17:27:19] [INFO] New client connected

11 [2026-02-12 17:27:31] [INFO] New node registered: node2

12 [2026-02-12 17:27:31] [INFO] New client connected

13 [2026-02-12 17:27:52] [INFO] Node-2 connected

14 [2026-02-12 17:27:52] [INFO] Session key generated for Node-2

15

client.loglogs

ECSBF_V1

> .vscode

> apps

> client

> engine

> include

> logs

client.log

engine.log

node1.log

node2.log

> src

Makefile

README.md

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

souvik@ray-device:/mnt/s/ECSBF_v1\$./apps/client/client

===== ECSBF NODE =====

1. Register

2. Login

3. Exit

Choose option: 1

Enter UserID: node2

Enter Password: 123

REGISTER_SUCCESS

Please login to continue.

===== ECSBF NODE =====

1. Register

2. Login

3. Exit

Choose option: 2

Enter UserID: node2

Enter Password: 123

LOGIN_SUCCESS

You can now chat.

Type 'exit' to quit.

Packet sent by Node-1: node1 hiii

node2 hello

OUTLINE

TIMELINE

PRACTICAL OUTCOMES – STEP 3 (Concurrent Processing & Logging)

- Engine handles multiple node signals using multi-threading.
- Sessions remain stable under simultaneous traffic load.
- Diagnostic logs capture INFO, WARNING, DEBUG, and FATAL events.
- System maintains low latency and real-time propagation.

Enterprise Concurrent Signal Broadcast Framework (ECSBF)

EXPLORER

... README.md engine.log client.log node1.log X

OPEN EDITORS

README.md engine.log logs client.log logs node1.log logs

ECSBF_V1

.vscode apps client engine include logs client.log engine.log node1.log node2.log src Makefile README.md

logs > node1.log

1 [2026-02-12 17:26:25] [INFO] Registration successful

2 [2026-02-12 17:26:45] [INFO] Login successful

3 [2026-02-12 17:28:52] [DEBUG] Encrypted Sent: +,7'wDEL;*,*,

4 [2026-02-12 17:28:52] [DEBUG] Decrypted Sent: node1 hiiii

5 [2026-02-12 17:29:09] [DEBUG] Encrypted Received: MAX"0)#+s6&<1t=2eem*'6otes+,6 fDEL# 5),

6 [2026-02-12 17:29:09] [DEBUG] Decrypted Received: Packet sent by Node-2: node2 hello

7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

===== ECSBF NODE =====

===== ECSBF NODE =====

1. Register

1. Register

2. Login

3. Exit

Choose option: 1

Enter UserID: node1

Enter Password: 123

REGISTER_SUCCESS

Please login to continue.

===== ECSBF NODE =====

1. Register

2. Login

3. Exit

Choose option: 2

Enter UserID: node1

Enter Password: 123

LOGIN_SUCCESS

You can now chat.

Type 'exit' to quit.

node1 hiiii

Packet sent by Node-2: node2 hello

EXPLORER

... README.md engine.log client.log node2.log X

OPEN EDITORS

README.md engine.log logs client.log logs node2.log logs

ECSBF_V1

.vscode apps client engine include logs client.log engine.log node1.log node2.log src Makefile README.md

logs > node2.log

1 [2026-02-12 17:27:31] [INFO] Registration successful

2 [2026-02-12 17:27:52] [INFO] Login successful

3 [2026-02-12 17:28:52] [DEBUG] Encrypted Received: MAX"0)#+s6&<1t=2eem*'6owest+,6 eDEL#,0,*

4 [2026-02-12 17:28:52] [DEBUG] Decrypted Received: Packet sent by Node-1: node1 hiiii

5 [2026-02-12 17:29:09] [DEBUG] Encrypted Sent: +,7'tDEL; />*

6 [2026-02-12 17:29:09] [DEBUG] Decrypted Sent: node2 hello

7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

3. Exit

Choose option: 1

Enter UserID: node2

Enter Password: 123

REGISTER_SUCCESS

Please login to continue.

===== ECSBF NODE =====

1. Register

2. Login

3. Exit

Choose option: 2

Enter UserID: node2

Enter Password: 123

LOGIN_SUCCESS

You can now chat.

Type 'exit' to quit.

Packet sent by Node-1: node1 hiiii

node2 hello

Enterprise Concurrent Signal Broadcast Framework (ECSBF)

EXPLORER

...

README.mdengine.log Xclient.lognode2.log

OPEN EDITORS

README.mdengine.log logsclient.log logsnode2.log logs

ECSBF_V1

> .vscode> apps> client> engine> include> logs

client.logengine.lognode1.lognode2.lognode3.log

> srcMakefileREADME.md

logs > engine.log

16 [2026-02-12 17:35:31] [INFO] New node registered: node3
17 [2026-02-12 17:35:31] [INFO] New client connected
18 [2026-02-12 17:35:50] [WARNING] Login failed for Node3
19 [2026-02-12 17:35:50] [INFO] New client connected
20

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

souvik@ray-device:/mnt/s/ECSBF_v1\$./apps/client/client

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option: 1
Enter UserID: node3
Enter Password: 123
REGISTER_SUCCESS
Please login to continue.

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option: 2
Enter UserID: Node3
Enter Password: 123
LOGIN_FAILED

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option:

EXPLORER

...

README.mdengine.lognode3.log Xclient.lognode2.log

OPEN EDITORS

README.mdengine.log logsnode3.log logsclient.log logsnode2.log logs

ECSBF...

> .vscode> apps> client> engine> include> logs

client.logengine.lognode1.lognode2.lognode3.log

> srcMakefileREADME.md

logs > node3.log

1 [2026-02-12 17:35:31] [INFO] Registration successful
2 [2026-02-12 17:35:50] [WARNING] Login failed
3

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

souvik@ray-device:/mnt/s/ECSBF_v1\$./apps/client/client

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option: 1
Enter UserID: node3
Enter Password: 123
REGISTER_SUCCESS
Please login to continue.

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option: 2
Enter UserID: Node3
Enter Password: 123
LOGIN_FAILED

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option:

PRACTICAL OUTCOMES – STEP 4 (Session Termination)

- Engine detects node disconnection or timeout event.
- Session is formally terminated to prevent orphan processes.
- Logging mechanism records session closure event.

Enterprise Concurrent Signal Broadcast Framework (ECSBF)

The screenshot shows a VS Code editor with the following components:

- EXPLORER:** Displays the project structure. The `logs` folder is expanded, showing `client.log`, `engine.log`, `node1.log`, `node2.log`, and `node3.log`. The `engine.log` file is selected.
- ENGINE.LOG:** Contains log entries for the engine. The visible entries are:

```
15 [2026-02-12 17:35:01] [INFO] New client connected
16 [2026-02-12 17:35:31] [INFO] New node registered: node3
17 [2026-02-12 17:35:31] [INFO] New client connected
18 [2026-02-12 17:35:50] [WARNING] Login failed for Node3
19 [2026-02-12 17:35:50] [INFO] New client connected
20 [2026-02-12 17:42:10] [INFO] Node-3 connected
21 [2026-02-12 17:42:10] [INFO] Session key generated for Node-3
22 [2026-02-12 17:42:24] [INFO] Client disconnected: node3
23
```
- TERMINAL:** Shows the output of the application. The visible output is:

```
Enter Password: 123
REGISTER_SUCCESS
Please login to continue.

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option: 2
Enter UserID: Node3
Enter Password: 123
LOGIN_FAILED

===== ECSBF NODE =====
1. Register
2. Login
3. Exit
Choose option: 2
Enter UserID: node3
Enter Password: 123
LOGIN_SUCCESS
You can now chat.
Type 'exit' to quit.
exit
souvik@ray-device:/mnt/s/ECSBF_v1$
```


LIMITATIONS:

- Requires stable TCP/IP network connectivity for optimal performance.
- Centralized Engine may become bottleneck under extreme scale.
- Currently backend-focused with no dedicated graphical UI.
- Performance depends on underlying hardware and OS thread management.
- Encryption overhead may slightly increase processing latency.

CONCLUSION:

- ECSBF provides secure, scalable, and concurrent signal broadcasting.
- Strong identity verification prevents unauthorized system access.
- Multi-threaded architecture ensures high-performance communication.
- Persistent session management enhances reliability and resilience.
- Modular and extensible design supports future enterprise enhancements.



**Thank
You!**

A	B	C	D	E
SOUVIK ROY				
SRS	Drafted Introduction, Purpose, and Project Scope sections defining overall system vision and objectives.			
DESIGN	Designed Core Engine architecture including Concurrency Engine and Signal Broadcaster modules.			
CODE	Implemented Engine Control Process, multi-threaded concurrency logic, and broadcast mechanism.			
CUT	Performed Engine-side unit testing and validated concurrent signal processing stability.			

J	K	L	M	N
ALISHA PARVEEN				
Prepared Non-Functional Requirements covering performance, security, scalability, and reliability aspects.				
Designed Secure Handshake, Encryption/Decryption flow, and Payload Handling structure.				
Implemented cryptographic modules for encryption, decryption, and secure payload transmission.				
Tested encryption integrity, payload validation, and secure transmission accuracy.				

F	G	H	I
KRITHIV DHARAN			
Documented System Features including Secure Registration and Concurrent Broadcasting requirements.			
Designed Global Identity Registry and Session Management workflow diagrams.			
Implemented Node connection handling, session establishment, and session restoration logic.			
Tested node connectivity, session recovery, and identity verification scenarios.			

O	P	Q	R	S	T
RITTIWIKA DATTA					
Compiled all external interface details, project assumptions, glossary terms, formatting updates, and finalized the complete ECSBF document.					
Designed the system architecture, logging structure, and complete message workflow for ECSBF.					
Implemented logging with FATAL, INFO, WARNING, and DEBUG levels, and handled proper session termination when clients disconnect.					
Integrated all ECSBF modules, tested full client–engine communication, fixed issues, and verified final system execution.					